

Natural Language Processing tasks

In this semester, there will be no exams within the scope of the course, instead, Python applied homeworks will be given. The student is expected to complete 10 tasks by selecting from the following Python coding tasks, and for each one, to prepare a YouTube video and explain these tasks in applicational.

Video presentation for each task should not exceed 5 minutes. In all of the video presentations, the student is required to make a video recording so that the identification can be made. In the video recording, the student should first introduce himself, then quickly tell the title of the project, if any, the dataset details, and the libraries he/she uses. He should then run the application and comment on the results in a few sentences.

Task 1. The results must be listed by applying at least 2 different tokenizers and 2 different lemmatizers representing the Tokenization and Lemmatization processes.

Task 2. Using N-gram analysis, list all 2-grams and 3-grams according to a certain threshold (eg, more than 5 in frequency) in a Turkish corpus. With these study, emphasize the importance of lemmatization for Turkish. Also, comment on what purpose this application can be used in the real world.

Task 3. By selecting 2 corpora prepared for different purposes, some basic features (unigram, bigram, trigram numbers, POS-Tag variety numbers etc.) should be compared and interpreted. The differences between the corpora and the relation between the preparation purposes of the corpora should be examined.

Task 4. A comparison will be made using different POS Taggers (at least two) on a corpus prepared for Turkish (the student will not prepare a new corpus, he/she must download it from internet, and the download link will be specified in the task). At least three words that POS Taggers labelled differently should be found and expressed.

Task 5. Basic semantic analysis of a Turkish corpus that will be downloaded from Internet will be done with statistical approaches. Accordingly, in the corpus;

- a list of all words (vocabulary) and their frequencies,
- a list of bigram and trigram tokens that have been used at least 5 times,
- the most similar two words using Latent Semantic Analysis should be determined.

Task 6. A very simple example should be prepared to show how to do word sense disambiguation using the LESK algorithm on a Turkish corpus. Here, the student can be flexible about corpus and glossary, in other words, a corpus with a few sentences and a dictionary with a few definitions can be prepared.

Task 7. First, low and high frequency words (for threshold = 5) should be determined in a Turkish collection. Words with low frequency will be assumed to have spelling errors. To solve this problem, a Lexical Similarity function will be used (student can find at internet or prepare) and it will be suggested that these erroneous assumed words can change to the most lexically similar word from high-frequency words. The code to be prepared must give a two-column list: low-frequency words in the first column and the most lexically similar high-frequency words in the second column.

Task 8. First of all, English word vectors calculated by any method (e.g. Word2Vec, GloVe or fastText) should be obtained from the internet. We must also have an English corpus. Using any Semantic Similarity method that the student chooses or creates, the most similar sentence in the corpus should be found for a sentence given externally.

Task 9. Student should prepare a Frequently Asked Questions chatbot based on one of BERT or ELMo or GPT-x models.

Task 10. Student should prepare a Machine Translation system based on one of BERT or ELMo or GPT-x models.

Task 11. Student should prepare a simple level Keyword Extraction or Document Summarization system.

Students can ask questions about homework just by attending live classes. Those who cannot attend the lesson because their lessons conflict, can write their message to the channel about the lesson via MS Teams. Emails about assignments sent outside of MS Teams will not be answered.

Apart from the headings above, students must prepare a FINAL task on a specific NLP problem (Presentation + Python). However, the title of the project you will choose for the FINAL task is first of all, you should ask the professor via MS Teams.